Large Language Models for Multimodal EHR Embedding and Clinical Prediction

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Abstract. Electronic Health Records (EHR) systems have become pervasive in modern healthcare settings. As the machine learning community focuses more and more on healthcare applications, an EHR with structured and unstructured patient data has become vital for inference tasks. In a decision support system, structured data facilitates easier retrieval and is useful, yet unstructured data contains crucial patient data embedded in clinical notes. The increasing volume and complexity of Electronic Health Records (EHRs) make it challenging for researchers and clinicians to manually extract clinically meaningful information. To better support clinical applications, there is a growing need for intelligent computational frameworks capable of jointly analyzing structured EHR data and unstructured clinical text. In this study, we propose an LLM-based framework for generating rich patient representations from EHR data. Our approach serializes structured tabular EHR features into natural language descriptions, enabling the use of pre-trained open-source Large Language Models (LLMs) to learn contextual and semantically meaningful representations. Furthermore, we evaluate the impact of incorporating clinical notes and free-text reports for downstream machine learning tasks, including clustering and ICU mortality prediction. Our findings indicate that our proposed approach outperforms traditional machine learning approaches for EHR representation. We evaluated the generated embeddings as both standalone and combined feature sets, demonstrating improved classification and clustering performance across multiple metrics, particularly ROC-AUC, which is critical in healthcare applications. Additionally, embedding space visualizations further highlights the robustness and effectiveness of the representations.

Keywords: Electronic Health record(EHR) · Clinical Notes · Large Language Models(LLMs) · Representation · Embeddings.

1 Introduction

Electronic Health Record (EHR) systems are ubiquitous in contemporary healthcare settings. They provide an integrated digital archive of patient health data gathered over time from different healthcare encounters [24]. Demographics, medical and family histories, test results from labs, radiological pictures, computed biosignals, and—most importantly—clinical notes written by doctors, nurses, and other members of the care team can all be found in EHRs [17].

EHRs are separate data sources that span several biological domains and scales and contain a broad range of data kinds, including numerical (like lab test results), categorical (like diagnosis codes, and drug types), and free-text data (clinical notes). This diversity makes a comprehensive understanding of patient health, outcomes, and treatments possible, but it also presents several challenges. EHR data is heterogeneous, which presents a challenge to traditional machine learning paradigms. Furthermore, the application of conventional feature engineering and one-hot encoding may obscure the underlying meanings in this data, leading to issues such as inadequate contextual understanding and sparse data representations. Addressing these challenges is crucial for making full use of the rich, yet complex, information EHR systems contain [38].

Considering the volume and variety of patient data contained in the EHR, which includes organized and unstructured data [7]. More intricate and extensive relationships, grouping, and classification problems are not well-served by traditional approaches. Consequently, it is imperative to create computational models that can handle EHR data efficiently. Applications for this might include helping with clinical diagnosis, selecting cohorts, determining medication efficacy, and improving clinical trial design. Increased accessibility to the electronic health record has provided healthcare systems with valuable and insightful data on patient medical histories and made its retrieval, storage, and accessibility easier in recent years [20]. With the growing attention that the machine learning community is paying to healthcare applications, this abundance of data makes EHR a vital resource for inference, insights, and performing underlying Machine Learning tasks [38].

The clinical notes are documentation of interactions with the healthcare system for a patient, encapsulating crucial information regarding diagnoses, treatments, care plans, and outcomes in rich, natural language [38]. A major drawback of these free text notes is that vital patient data is buried in massive amounts of unorganized clinical notes, making it difficult for clinicians to quickly and effectively look through and extract pertinent information [17]. There has been a recent focus on integrating state-of-the-art foundation models, such as Large Language Models (LLMs) [42], with both conventional and innovative machine learning approaches being used to help with specific diagnosis and prognosis tasks [37]. These sophisticated models, which have been pre-trained on large and varied textual corpora, provide a comprehensive comprehension across multiple domains, increasing their versatility and efficacy for a range of challenging tasks [38]. Notably, they frequently exhibit state-of-the-art performance when directly applied to numerous classification problems thanks to their capacity to produce high-fidelity latent representations.

The integration of Foundation Models and their autoregressive nature and capacity for zero-shot learning drive ongoing advancements in language processing capabilities [31]. Recent developments in large language models (LLMs) have suggested that they could transform medical applications, as demonstrated by ChatGPT and LLaMA [39, 41]. We must assess LLMs in the wild, that is, on actual data under realistic circumstances, to comprehend their use in real-world

scenarios. The use of LLMs, however, is not without its difficulties. For example, ChatGPT and other likewise numerous LLMs are located in remote data centers and are therefore ineligible to handle personal healthcare data due to the sensitivity of the information [3]. The generability of these widely used models, which are intended for text and natural language, is another problem. In clinical and EHR settings, we aim to evaluate the utility and robustness of these general-purpose models while ensuring that privacy concerns are effectively addressed.

With LLM's remarkable natural language synthesis and generating capabilities, we hypothesize that LLM could be applied to effectively generate embeddings from EHR data using its structured and unstructured patient information. In this work, we present a method to generate embeddings using open-source foundation models LLM such as LLaMA2, LLaMA3, and Mistral. We have constructed a cohort dataset using the MIMIC-IV [21, 22, 13] database, specifically the hospital module. Each patient case contains the filtered and curated information necessary to find the survival chances of the patient and can be used to test the feature extraction capabilities of these LLM models using real-world data [16, 15]. The tabular data is serialized to a string in natural language and then prompted to the large language model to generate numerical embeddings [18, 31]. We evaluate our proposed methods for the task of mortality prediction and clustering to get insights about patients and correlations for ICU admittance. In this work, our contributions can be summarized as follows:

1. We present a pipeline using the local LLMs Llama 2, Llama3, and Mistral models for extracting quantitative information from the clinical text, along with other features.
2. We use two embedding generation techniques:
   - We serialized the tabular data to form a free text using the features, and
   - In addition, we used clinical notes from the MIMIC-IV dataset as input to generate embeddings.
3. We evaluate these embeddings by performing the clustering and binary classification (patient survival prediction) task.
4. Pre-trained LLMs are used in a local setting to address privacy concerns of EHR data. Thus none of the data is being transmitted to servers.
5. We use t-Sne visualization to evaluate the embeddings subjectively and strengthen the predictive robustness of the proposed method.
6. Results revealed the superior performance of the proposed embeddings, showcasing improvements in AUC-ROC scores and other metrics.

2 Related Work

The increasing availability of Electronic Health Records (EHRs) has improved patient care while enabling data-driven healthcare research [37]. In intensive care units (ICUs), clinical data has been widely used to assess patient health and predict critical outcomes such as mortality [23]. Authors in [19] used a logistic regression model. The prediction of in-hospital mortality for patients who stayed in the ICU for at least 48 hours [8]. It is estimated that 80% of clinical data exists

in an unstructured format [25]. Natural language has not been widely used as a source to extract structured information [29]. Transforming unstructured clinical data into formats suitable for downstream tasks can improve patient outcomes and enable valuable scientific insights [35].

ML models have shown potential on EHR data to predict a wide array of clinical and operational outcomes, to risk-stratify patients and inform treatment decisions such as predicting mortality [23] and length of stay (LoS) [2]. Techniques for serializing tabular data into textual formats have enabled better integration of structured data with language models, supporting advanced machine learning applications [18]. However, limited annotated clinical data remains a major bottleneck for clinical NLP systems [38]. This capability has been demonstrated through prompt-based text-to-text frameworks that use serialized tabular data to leverage LLM knowledge for zero-shot and few-shot learning tasks [18].

A more recent approach involves constructing patient summaries directly from tabular data using LLMs for natural integration into future NLP tasks, highlighted by [18, 11]. Authors in [7] use pre-trained language models like LLaMA 2 to derive contextual representations from EHR and criteria inputs in the domain of clinical trial patient recruitment. Authors in [41] propose Me-LLaMA, developed through continual pre-training and instruction tuning of LLaMA2 using large medical datasets. It stands out as one of the largest and most effective open-source LLMs for medical applications, bridging the gap between general language models and medical-specific tasks. Authors in [27] propose a Multiple Embedding Model for EHR (MEME), an approach that serializes multimodal EHR tabular data into text using "pseudo-notes", mimicking clinical text generation. A few research works such as PMC-LLaMA [40] and Meditron [5] only focus on the biomedical domain and evaluate only the question-answering (QA) task. Very few of them used clinical data and evaluated clinical tasks where GatorTronGPT [33] and Clinical-LLaMA [12] are two exceptions.

Clinical LLMs are often fine-tuned with clinical data from an LLM that is already pretrained with datasets that encompass broader topics. BioBERT [26], PubMedBERT [14], Bio+ClinicalBERT [1], BlueBERT [34], and Clinical-T5 [28]. Most existing studies rely on full model fine-tuning, which has become computationally expensive and impractical with large-scale LLMs. Foundation Models demonstrate strong generalization across diverse tasks, particularly in data cleaning and integration, though their effectiveness is often tied to prompt-based settings [31]. Recent research has increasingly explored the use of AI and LLMs to enhance clinical decision-making from clinical notes, including conversational systems for intuitive querying and insight retrieval [10, 6].

3 Proposed Approach

The clinical data were extracted from the MIMIC-IV hospital module and preprocessed using extraction pipelines (available in the published version). The curated cohort consisted of ICU patients satisfying the inclusion criteria described in the 4.4 section. Structured patient information, including demographic vari-

ables, vital signs, and laboratory measurements, was transformed into a feature matrix for downstream modeling. Figure 1 illustrates the overall framework of the proposed approach for generating patient representations using both structured clinical variables and unstructured discharge notes from the MIMIC-IV dataset. The framework integrates tabular clinical information with Large Language Model (LLM)-based textual embeddings to improve downstream ICU mortality prediction and clustering performance.

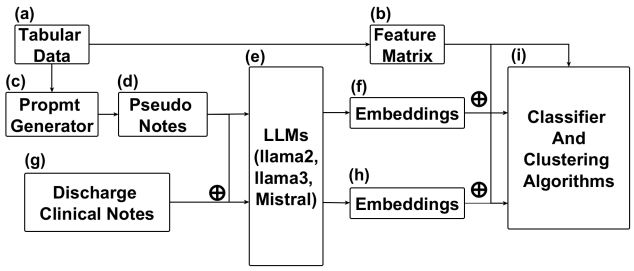


Fig. 1: Proposed pipeline Workflow

As shown in Figure 1(a)–(d), the proposed pipeline first converts structured tabular patient data into descriptive pseudo-clinical notes using a prompt generation module. The prompt generator transforms numerical and categorical patient variables into natural language summaries describing the patient's physiological condition. These generated pseudo-notes aim to bridge the semantic gap between structured EHR features and language-based representation learning. The generated pseudo-notes are then provided as input to multiple open-source LLMs, including *Llama-2*, *Llama-3*, and *Mistral*, as illustrated in Figure 1(e). The LLMs are used to generate dense embedding representations capturing contextual clinical semantics from the pseudo-notes. In parallel, original discharge clinical notes extracted from MIMIC-IV are also processed through the same LLM-based embedding pipeline to obtain textual embeddings derived from real clinician documentation, as shown in Figure 1(g)–(h). Simultaneously, the selected structured variables are normalized and represented as a traditional feature matrix, shown in Figure 1(b). The resulting structured feature vectors are combined with the embedding representations generated from pseudo-notes and discharge notes using feature fusion operations. This multimodal integration enables the framework to leverage both explicit clinical measurements and latent semantic information learned from textual narratives.

Finally, the fused representations are provided to downstream machine learning models and clustering algorithms for ICU mortality prediction and patient representation analysis, as shown in Figure 1(i). Multiple classification algorithms were evaluated to assess predictive performance, while clustering techniques were used to analyze the quality and separability of the learned patient embeddings. The proposed framework enables effective integration of structured

EHR features with LLM-generated representations, facilitating improved clinical representation learning and downstream predictive modeling tasks.

4 Experimental Setup

In this section, we describe the classifiers, regression models, and evaluation metrics used for the computation of results. We also describe the baseline models used for the comparison of classification and regression results. We then provide the detail regarding the dataset used for experimentation followed by the visual analysis of data to understand the distribution of target labels among different embedding methods.

4.1 Classification Evaluation Metrics

In our classification task, we utilize a variety of linear and non-linear classifiers such as SVM, Naive Bayes (NB), Multi-Layer Perceptron (MLP), K Nearest Neighbors (KNN), Random Forest (RF), Logistic Regression (LR), and Decision Tree (DT). Our evaluation metrics include average accuracy, precision, recall, weighted F1, macro F1, ROC-AUC, and classifier training runtime. We split our data into random training and test sets with a $70 - 30\%$ split, and we repeat our experiments 5 times to obtain average results. For hyperparameter tuning, we reserve 70% for train validation (10% of which is validation) and 30% as a held-out test set. To ensure reproducibility, we provide access to our code and pre-processed dataset online.

4.2 Clustering Evaluation Metrics

In this section, we describe the internal clustering evaluation measures used to assess the quality of clustering.

Silhouette Coefficient [36]. Given a feature vector, the silhouette coefficient computes how similar the feature vector is to its own cluster (cohesion) compared to other nearest clusters (separation) [32]. Its score ranges between $[-1, 1]$, where 1 means the best possible clustering and -1 means the worst possible clustering. Score close to 0 indicate that the sample is on or near the decision boundary between two neighboring clusters, whereas negative values indicate that the sample is likely to be assigned to an incorrect cluster.

Calinski-Harabasz Score [4]. The Calinski-Harabasz (CH) score, also known as the Variance Ratio Criterion, is a measured as the ratio of the sum of between-cluster dispersion to within-cluster dispersion. It evaluates the validity of a clustering based on the within-cluster and between-clusters dispersion of each object with respect to each cluster (based on sum of squared distances) [32]. A higher score denotes better defined and more compact clusters. The score is higher when clusters are well-separated and have high intra-cluster similarity.

Davies-Bouldin Score [9]. Given a feature vector, the Davies-Bouldin score computes the ratio of within-cluster to between-cluster distances [32]. It measures the ratio of the sum of the within-cluster dispersions to the distance between cluster centroids. A smaller score denotes groups are well separated, and better clustering. It means clusters are more compact (small intra-cluster dispersion) and well-separated from each other (large inter-cluster distances). On theotherside higher score denotes clusters are either too dispersed within themselves or overlap significantly with neighboring clusters.

4.3 LLM Models

In this section, we describe different LLM models used for generating embeddings for comparative evaluation. To generate semantic patient representations from pseudo-clinical notes and discharge summaries, we utilized three open-source Large Language Models (LLMs): *Llama-2*, *Llama-3*, and *Mistral-7B*. These models were selected due to their strong language understanding capabilities, open-source availability, and efficient deployment in local clinical research environments. All models were executed locally to ensure data privacy and compliance with healthcare data handling requirements.

Llama-2: Developed by Meta Platforms, it is a decoder-only transformer-based autoregressive language model built upon the standard Transformer architecture. The model was trained on approximately 2 trillion tokens collected from publicly available sources and further fine-tuned using more than 1 million human preference annotations through reinforcement learning from human feedback (RLHF).

Llama-3: It is the next-generation transformer-based LLM released by Meta, designed with improved tokenizer efficiency, enhanced contextual reasoning, and larger-scale pretraining. The model introduces a tokenizer with a vocabulary size of 128K tokens, enabling more efficient encoding of clinical and biomedical terminology. Llama-3 was pretrained on over 15 trillion tokens from diverse publicly available datasets.

Mistral-7B: Developed by Mistral AI, is a lightweight yet high-performing decoder-only Transformer model containing approximately 7.3 billion parameters. Despite its smaller size, Mistral-7B achieves competitive performance against LLMs through architectural optimizations such as grouped-query attention (GQA) and sliding window attention (SWA), which significantly improve memory efficiency and long-context processing.

Embedding vectors for all models were generated using locally deployed inference endpoints through the embedding API interface. Each pseudo-note and discharge note was converted into dense vector representations using the corresponding LLM embedding pipeline. These embeddings were subsequently fused with structured feature matrices to support downstream classification and clustering experiments for ICU mortality prediction.

4.4 Dataset

Cohort Construction and Dataset Description: We utilized the MIMIC-IV [21, 22, 13] dataset to construct the study cohort for ICU mortality prediction. Data were curated from multiple clinical tables, including chart events, laboratory measurements, outpatient records, and discharge notes. To ensure data quality and consistency, only the first 24 hours of ICU data were considered for each patient admission. Additionally, patients younger than 15 years of age and ICU stays shorter than 12 hours were excluded from the analysis. After applying the initial filtering criteria, a total of 45,919 ICU records were retained. Subsequently, discharge notes were used to further refine the cohort by selecting patients with note entries documented within 48 hours of ICU admission. The final study cohort consisted of 6,818 patients.

Demographic Statistics: The final cohort included both male and female ICU patients with a relatively balanced gender distribution. Specifically, the dataset contained 3,614 female patients and 3,204 male patients. Mortality distribution analysis showed that 5,596 patients survived ICU admission, whereas 1,222 patients experienced in-hospital mortality.

Feature Statistics: A total of 18 clinically relevant features were extracted from multiple tables in the MIMIC-IV database to support downstream predictive modeling and embedding evaluation tasks. These features were selected based on their clinical importance in assessing patient severity, physiological stability, and ICU outcomes. The extracted variables capture diverse aspects of patient health, including demographic information, neurological assessments, vital signs, and laboratory measurements.

The prediction target for all experiments was defined as **ICU mortality**, where each patient record was categorized into either survival or mortality outcome classes. Analysis of cohort statistics revealed a moderate class imbalance, with survival cases significantly outnumbering mortality cases.

4.5 Data Visualization

In order to assess whether various embedding techniques are maintaining the structure of the data, we employed the t-distributed Stochastic Neighbour Embedding (t-SNE) [30] algorithm to generate 2-dimensional representations of the embeddings for visual inspection. The t-SNE visualizations of patient embeddings were generated under two experimental settings: embeddings generated without clinical notes Figure 2 and embeddings generated with both prompts and discharge clinical notes Figure 3. The visualization results provide additional insights into the clustering behavior and separability of patient representations generated by different embedding approaches.

As we can see in Figure 2, the embeddings generated without clinical notes produce relatively more distinguishable and well-separated cluster structures, particularly for the Llama-3 and FM + Llama-3 representations as shown in

Figure 2 (b),(c), and (d), in contrast to the classical feature matrix representation Figure 2(a). The classical feature matrix produced overlapping clusters with irregular boundaries, indicating limited separability in the structured feature space. These results suggest that LLM embeddings capture richer semantic and contextual clinical relationships that are difficult to model using traditional tabular features alone.

Although the embeddings generated without clinical notes Figure 2 produce relatively more distinguishable and well-separated cluster structures in comparison to embeddings generated with clinical notes 3. A possible explanation is that discharge clinical notes contain semantically rich and highly detailed medical narratives describing similar clinical conditions, treatments, and disease progression patterns across different patient groups. Although the inclusion of these notes enhances the semantic quality of the learned embeddings, it can also create shared latent representations among clinically similar patients, resulting in increased overlap in the low-dimensional t-SNE space. Despite the reduced visual separability, the additional contextual information improves downstream predictive performance, which is reflected in the classification results where Llama-3 achieves significantly better performance when clinical notes are incorporated.

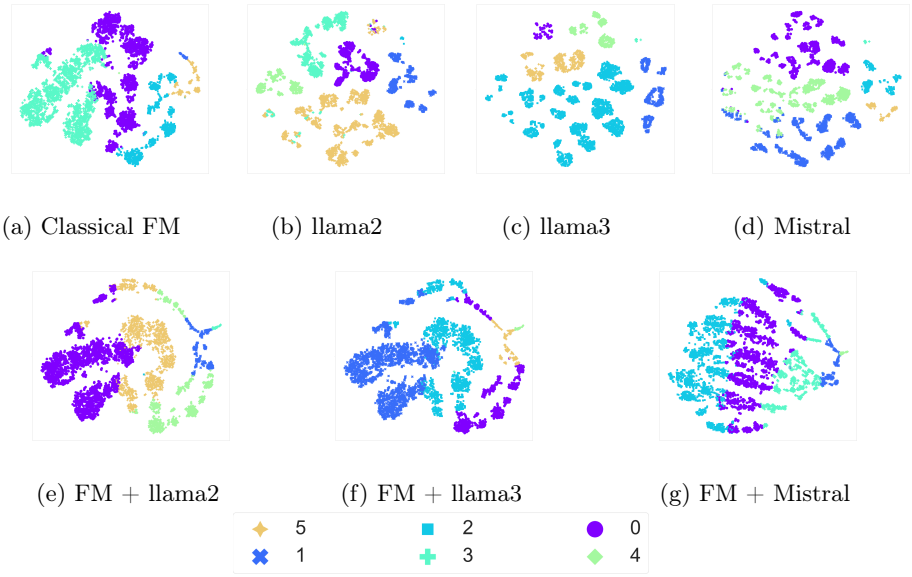
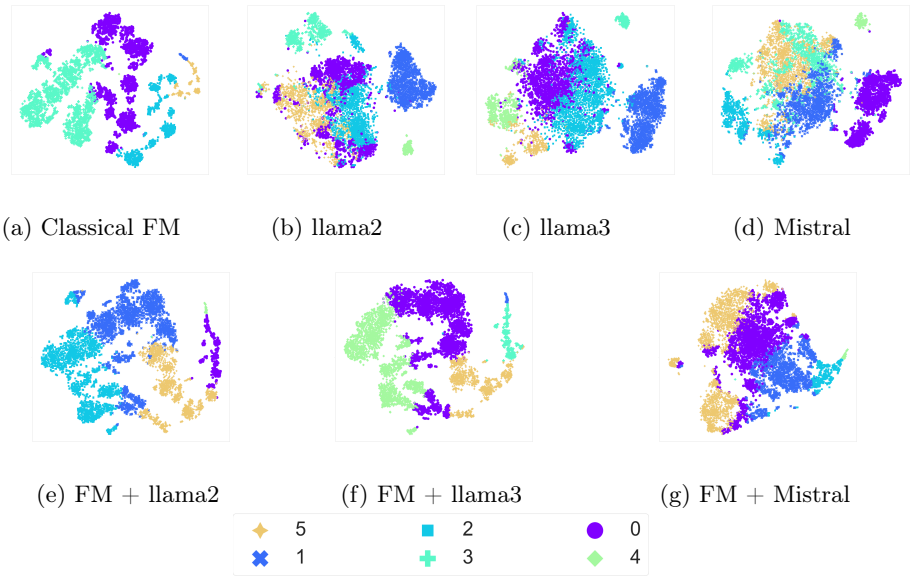


Fig. 2: t-SNE visualization of different embeddings from **MIMIC-IV dataset** (without clinical notes) for kmeans clustering. The Figure is best seen in color.



(a) Classical FM (b) llama2 (c) llama3 (d) Mistral

(e) FM + llama2 (f) FM + llama3 (g) FM + Mistral

Fig. 3: t-SNE visualization of different embeddings from **MIMIC-IV dataset** (with clinical notes) for kmeans clustering. The Figure is best seen in color.

5 Results And Discussion

5.1 Classification Performance Analysis

We evaluated the proposed framework using multiple machine learning classifiers on both structured feature matrices and LLM-generated embeddings. The experiments were conducted under two settings: (1) embeddings generated using serialized prompts only, as shown in Table 3, and (2) embeddings generated using prompts combined with discharge clinical notes, results shown in Table 4.

The experimental results demonstrate that embeddings generated solely from serialized prompts provide comparable classification performance over the classical feature matrix baseline. Although the prompt-based embeddings captured semantic relationships from structured EHR variables, the absence of rich clinical narratives restricted the overall discriminative capability of the learned patient representations. In contrast, incorporating discharge clinical notes substantially improved predictive performance across nearly all evaluation metrics, including accuracy, precision, recall, weighted F1-score, macro F1-score, and ROC-AUC. Among the evaluated models, Llama-3 embeddings consistently achieved the strongest overall performance, outperforming both the baseline feature matrix and other LLM variants. As we can see in the Table 4 the Random Forest classifier using Llama-3 embeddings achieved the highest classification accuracy (95.9%) and the same with all other metrics, while also demonstrating strong ROC-AUC scores (73.2%) with NB classifier.

The improvement in ROC-AUC indicates that the semantic information contained in clinical notes significantly enhances the separability between mortality and survival classes. Compared to the classical feature matrix baseline, the proposed LLM-based framework achieved approximately 7 to 8% improvement in ROC-AUC performance. These findings suggest that contextual clinical narratives contain important latent information that may not be fully represented through structured ICU measurements alone.

Additionally, the fusion of structured features with LLM embeddings demonstrated stable and competitive performance across multiple classifiers. This indicates that combining explicit physiological measurements with semantic textual representations provides complementary information for downstream predictive modeling tasks. From a computational perspective, larger embedding models generally require longer training and inference times. However, the increased computational overhead was justified by the significant gains in predictive performance and representation quality observed in the experiments.

5.2 Clustering Performance Analysis

We conducted unsupervised clustering experiments using *k-means* and *k-modes* clustering algorithms. Clustering quality was assessed using Silhouette Coefficient, Calinski-Harabasz Score, Davies-Bouldin Score, and clustering runtime. The optimal number of clusters was determined using the elbow method by analyzing the within-cluster sum of squares (WCSS) across different cluster values.

The clustering results in Table 1 showed that prompt-only embeddings (without clinical notes) generated by LLMs improved cluster separability compared to the classical feature matrix baseline. In particular, Llama-3 embeddings achieved the highest Calinski-Harabasz scores, indicating stronger cluster compactness and better inter-cluster separation. Similarly, Llama-2 demonstrated competitive Silhouette Coefficient values, suggesting improved representation consistency within clusters. Table 2 shows the clustering performance on the MIMIC-IV dataset with clinical notes. Standalone LLM embeddings showed lower clustering separability and increased overlap, which is consistent with the t-SNE visualizations. A possible explanation is that discharge clinical notes contain semantically rich and detailed narratives describing similar clinical conditions across patient groups, leading to shared latent representations and overlapping clusters in the embedding space. Despite this reduced visual separability, the inclusion of clinical notes significantly improved downstream classification performance, particularly for Llama3-based models.

6 Conclusion

In this work, we propose a multimodal framework that combines structured EHR features with free-text discharge clinical notes to generate patient embeddings using multiple open-source LLMs, including Llama-2, Llama-3, and Mistral. Experimental results demonstrated that prompt-based embeddings alone achieved

Embeddings	k-means				k-modes			
	Silhouette Coefficient	Calinski Harabasz Score	Davies-Bouldin Score	Clustering Runtime (Sec.)	Silhouette Coefficient	Calinski Harabasz Score	Davies-Bouldin Score	Clustering Runtime (Sec.)
Classical FM	0.4174	9263.8019	**0.5923**	**0.5808**	**-0.0726**	2.6754	40.5961	**0.9251**
llama2	**0.4193**	10565.1773	0.9113	7.2313	-0.3551	1.6740	7.2633	95.1870
llama3	0.3829	**13788.0223**	0.8945	7.9683	-0.3686	2.2413	6.8917	105.6733
Mistral	0.2973	5189.2681	1.1187	8.5395	-0.3080	1.3991	**5.6648**	102.5672
FM + llama2	0.3781	8487.3842	0.6490	2.9127	-0.0794	**6.4403**	42.2409	158.6200
FM + llama3	0.3845	8365.1290	0.6460	2.6659	-0.0858	5.4242	30.6139	135.2033
FM + Mistral	0.3046	5770.7152	0.8003	2.9130	-0.0753	5.2045	29.8278	139.2221

Table 1: Clustering results for **MIMIC-IV dataset** (Prompts without clinical notes) on different embeddings. The best values are shown in bold.

Embeddings	k-means				k-modes			
	Silhouette Coefficient	Calinski Harabasz Score	Davies-Bouldin Score	Clustering Runtime (Sec.)	Silhouette Coefficient	Calinski Harabasz Score	Davies-Bouldin Score	Clustering Runtime (Sec.)
Classical FM	**0.4174**	**9263.8019**	**0.5923**	**0.5808**	**-0.0726**	**2.6754**	40.5961	**0.9251**
llama2	0.1239	663.6874	2.3269	9.0099	-0.0734	1.0925	**5.4703**	102.5305
llama3	0.1151	750.6148	2.4717	9.5323	-0.0876	0.9613	6.1446	111.1882
Mistral	0.1037	562.3429	2.5766	11.8593	-0.0739	0.8779	5.5696	100.9482
FM + llama2	0.3330	7764.4627	0.7231	2.7757	-0.0867	5.1294	28.9581	143.8941
FM + llama3	0.3365	7504.6235	0.7141	3.0744	-0.0851	5.5773	28.9350	152.9987
FM + Mistral	0.1582	3211.0147	1.2475	4.4300	-0.0564	5.3170	27.5824	135.7623

Table 2: Clustering results for **MIMIC-IV dataset** (with clinical notes) on different embeddings. The best values are shown in bold.

performance comparable to traditional structured feature matrix baselines, indicating that LLMs can effectively capture semantic relationships from serialized EHR data. However, incorporating discharge clinical notes significantly improved predictive performance across all evaluation metrics, including accuracy, F1-score, and ROC-AUC. Among all evaluated models, Llama-3 consistently achieved the strongest performance, demonstrating the effectiveness of advanced transformer-based embeddings for predictive clinical modeling. In general, the findings suggest that the combination of structured clinical data with unstructured clinical notes enables more meaningful patient representation learning than relying on tabular features alone. The proposed framework demonstrates the potential of LLM-based embedding methods for clinical prediction, patient stratification, and healthcare representation learning tasks. In future work, it could be interesting to incorporate larger biomedical foundation models, temporal patient trajectories, and domain-specific clinical LLMs to further improve representation quality and predictive performance.

Embed. Method	ML Algo.	Acc.	Prec.	Recall	F1 weigh.	F1 Macro	ROC-AUC	Train. run-time (sec.)
Classical FM	SVM	0.887	0.879	0.887	0.881	0.782	0.628	0.581
	NB	0.878	0.880	0.878	0.879	0.791	0.646	**0.003**
	MLP	0.885	0.886	0.885	0.886	0.802	0.651	4.177
	KNN	0.861	0.847	0.861	0.848	0.717	0.594	0.148
	RF	**0.907**	**0.904**	**0.907**	**0.905**	**0.831**	**0.659**	0.758
	LR	0.883	0.875	0.883	0.877	0.775	0.625	0.010
	DT	0.856	0.857	0.856	0.856	0.751	0.626	0.046
llama2	SVM	0.856	0.858	0.856	0.857	0.757	0.630	22.753
	NB	0.312	0.807	0.312	0.299	0.311	0.530	0.236
	MLP	0.871	0.869	0.871	0.870	0.775	0.634	8.263
	KNN	0.855	0.839	0.855	0.840	0.704	0.588	0.808
	RF	0.862	0.849	0.862	0.848	0.719	0.594	56.210
	LR	0.879	0.877	0.879	0.878	0.789	0.641	37.492
	DT	0.815	0.817	0.815	0.816	0.687	0.595	64.377
llama3	SVM	0.848	0.848	0.848	0.848	0.744	0.622	26.112
	NB	0.309	0.818	0.309	0.290	0.307	0.532	0.296
	MLP	0.864	0.868	0.864	0.865	0.776	0.642	9.124
	KNN	0.854	0.840	0.854	0.842	0.716	0.595	0.959
	RF	0.858	0.844	0.858	0.844	0.717	0.594	55.605
	LR	0.870	0.866	0.870	0.867	0.773	0.632	36.944
	DT	0.810	0.809	0.810	0.809	0.677	0.588	62.055
Mistral	SVM	0.854	0.856	0.854	0.855	0.752	0.627	21.979
	NB	0.330	0.814	0.330	0.327	0.330	0.536	0.246
	MLP	0.872	0.872	0.872	0.871	0.778	0.639	6.585
	KNN	0.861	0.850	0.861	0.853	0.734	0.606	0.832
	RF	0.865	0.853	0.865	0.855	0.737	0.606	61.383
	LR	0.877	0.876	0.877	0.876	0.786	0.641	35.168
	DT	0.811	0.817	0.811	0.814	0.686	0.596	77.359
FM + llama2	SVM	0.851	0.853	0.851	0.852	0.743	0.623	23.578
	NB	0.302	0.831	0.302	0.286	0.301	0.533	0.340
	MLP	0.880	0.882	0.880	0.881	0.794	0.650	10.580
	KNN	0.863	0.849	0.863	0.850	0.720	0.595	0.666
	RF	0.874	0.863	0.874	0.862	0.741	0.604	53.681
	LR	0.878	0.875	0.878	0.876	0.780	0.637	33.787
	DT	0.842	0.844	0.842	0.843	0.728	0.616	45.952
FM + llama3	SVM	0.854	0.857	0.854	0.855	0.754	0.630	21.485
	NB	0.304	0.826	0.304	0.286	0.303	0.533	0.370
	MLP	0.878	0.878	0.878	0.878	0.789	0.644	10.511
	KNN	0.865	0.852	0.865	0.852	0.726	0.598	0.609
	RF	0.870	0.859	0.870	0.859	0.741	0.606	48.526
	LR	0.875	0.874	0.875	0.874	0.782	0.639	41.485
	DT	0.848	0.849	0.848	0.848	0.740	0.621	46.173
FM + mistral	SVM	0.859	0.860	0.859	0.859	0.761	0.632	20.953
	NB	0.331	0.817	0.331	0.328	0.331	0.537	0.368
	MLP	0.880	0.882	0.880	0.881	0.798	0.651	12.573
	KNN	0.868	0.856	0.868	0.856	0.736	0.602	0.640
	RF	0.874	0.864	0.874	0.865	0.754	0.613	57.214
	LR	0.880	0.877	0.880	0.878	0.790	0.641	34.798
	DT	0.849	0.851	0.849	0.850	0.744	0.623	38.530

Table 3: Classification results for different embeddings for MIMIC-IV dataset (without clinical notes). The top 5% values for each metric are underlined.

Embed. Method	ML Algo.	Acc.	Prec.	Recall	F1 weigh.	F1 Macro	ROC-AUC	Train. run-time (sec.)
Classical FM	SVM	0.887	0.879	0.887	0.881	0.782	0.628	0.581
	NB	0.878	0.880	0.878	0.879	0.791	0.646	0.003
	MLP	0.885	0.886	0.885	0.886	0.802	0.651	4.177
	KNN	0.861	0.847	0.861	0.848	0.717	0.594	0.148
	RF	0.907	0.904	0.907	0.905	0.831	0.659	0.758
	LR	0.883	0.875	0.883	0.877	0.775	0.625	0.010
	DT	0.856	0.857	0.856	0.856	0.751	0.626	0.046
llama2	SVM	0.908	0.903	0.908	0.905	0.581	0.539	5.295
	NB	0.654	0.909	0.654	0.743	0.480	0.566	0.139
	MLP	0.926	0.903	0.926	0.912	0.562	0.524	2.928
	KNN	0.935	0.882	0.935	0.907	0.486	0.500	0.439
	RF	0.938	0.880	0.938	0.908	0.484	0.500	40.605
	LR	0.914	0.901	0.914	0.907	0.569	0.531	10.856
	DT	0.886	0.890	0.886	0.888	0.527	0.515	46.099
llama3	SVM	0.942	0.942	0.942	0.941	0.901	0.699	8.599
	NB	0.952	0.960	0.952	0.954	0.925	**0.732**	0.241
	MLP	0.949	0.950	0.949	0.950	0.915	0.711	4.219
	KNN	0.955	0.960	0.955	0.956	0.928	0.728	0.813
	RF	**0.959**	**0.963**	**0.959**	**0.960**	**0.934**	0.729	31.146
	LR	0.949	0.951	0.949	0.950	0.916	0.714	11.387
	DT	0.928	0.927	0.928	0.927	0.876	0.686	28.706
Mistral	SVM	0.937	0.938	0.937	0.937	0.893	0.699	8.744
	NB	0.946	0.957	0.946	0.948	0.915	0.731	0.239
	MLP	0.946	0.949	0.946	0.947	0.912	0.714	4.020
	KNN	0.950	0.957	0.950	0.952	0.921	0.726	0.822
	RF	0.955	0.960	0.955	0.956	0.927	0.729	33.436
	LR	0.945	0.948	0.945	0.946	0.910	0.714	11.703
	DT	0.921	0.920	0.921	0.921	0.863	0.680	26.621
FM + llama2	SVM	0.939	0.939	0.939	0.939	0.897	0.699	9.124
	NB	0.951	0.959	0.951	0.953	0.924	0.731	0.327
	MLP	0.947	0.948	0.947	0.947	0.912	0.710	4.913
	KNN	0.938	0.938	0.938	0.938	0.895	0.699	0.632
	RF	0.956	0.960	0.956	0.957	0.930	0.728	34.540
	LR	0.947	0.949	0.947	0.948	0.913	0.713	12.410
	DT	0.927	0.926	0.927	0.926	0.875	0.686	24.488
FM + llama3	SVM	0.942	0.944	0.942	0.943	0.901	0.705	8.259
	NB	0.948	0.958	0.948	0.951	0.917	0.731	0.340
	MLP	0.944	0.946	0.944	0.944	0.904	0.707	6.044
	KNN	0.944	0.947	0.944	0.945	0.906	0.712	0.634
	RF	0.955	0.960	0.955	0.956	0.926	0.728	31.343
	LR	0.946	0.950	0.946	0.948	0.910	0.716	11.876
	DT	0.925	0.925	0.925	0.925	0.868	0.684	25.837
FM + mistral	SVM	0.936	0.937	0.936	0.936	0.892	0.699	8.176
	NB	0.943	0.955	0.943	0.946	0.913	0.730	0.319
	MLP	0.944	0.947	0.944	0.945	0.908	0.713	5.160
	KNN	0.943	0.949	0.943	0.945	0.909	0.719	0.649
	RF	0.952	0.959	0.952	0.954	0.924	0.729	34.519
	LR	0.939	0.943	0.939	0.940	0.900	0.710	10.605
	DT	0.920	0.921	0.920	0.921	0.865	0.684	26.916

Table 4: Classification results for different embeddings for the MIMIC-IV (with clinical notes) dataset. The top 5% values for each metric are underlined.

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Embed. Method	ML Algo.	Acc.	Prec.	Recall	F1 weigh.	F1 Macro	ROC-AUC	Train. runtime (sec.)
Classical FM	SVM	0.00862	0.00951	0.00862	0.00897	0.01401	0.00617	0.04517
	NB	0.01033	0.00665	0.01033	0.00865	0.01338	0.00538	0.00121
	MLP	0.01139	0.00950	0.01139	0.01036	0.01623	0.00621	1.27089
	KNN	0.00887	0.01051	0.00887	0.00900	0.01667	0.00708	0.00493
	RF	0.01090	0.01200	0.01090	0.01165	0.02146	0.01174	0.02656
	LR	0.00900	0.01059	0.00900	0.01023	0.01889	0.00954	0.00307
	DT	0.00672	0.00704	0.00672	0.00679	0.01721	0.00924	0.00322
llama2	SVM	0.00550	0.00377	0.00550	0.00447	0.00957	0.00357	0.71160
	NB	0.00711	0.01278	0.00711	0.00986	0.00726	0.00257	0.02867
	MLP	0.00631	0.00657	0.00631	0.00627	0.01354	0.00824	1.51726
	KNN	0.00466	0.00661	0.00466	0.00706	0.01611	0.00804	0.02078
	RF	0.00453	0.00600	0.00453	0.00570	0.01396	0.00683	5.78057
	LR	0.00094	0.00257	0.00094	0.00188	0.00387	0.00342	8.30975
	DT	0.00651	0.00356	0.00651	0.00448	0.00715	0.00331	11.63862
llama3	SVM	0.00617	0.00750	0.00617	0.00674	0.00931	0.00501	2.19428
	NB	0.01364	0.01033	0.01364	0.01386	0.01348	0.00439	0.07191
	MLP	0.00879	0.00784	0.00879	0.00802	0.01131	0.00928	1.86305
	KNN	0.00761	0.00848	0.00761	0.00879	0.00903	0.00424	0.09842
	RF	0.00526	0.00553	0.00526	0.00658	0.00328	0.00154	3.51304
	LR	0.00318	0.00441	0.00318	0.00388	0.00848	0.00446	8.45684
	DT	0.01072	0.01017	0.01072	0.01037	0.01661	0.00807	13.14359
Mistral	SVM	0.00633	0.00682	0.00633	0.00597	0.01129	0.00624	0.94418
	NB	0.01272	0.01805	0.01272	0.01249	0.01273	0.00357	0.01589
	MLP	0.00648	0.01137	0.00648	0.00884	0.01788	0.01538	0.72813
	KNN	0.00756	0.00848	0.00756	0.00865	0.00868	0.00443	0.03955
	RF	0.00820	0.00985	0.00820	0.01063	0.01064	0.00672	3.24269
	LR	0.00324	0.00571	0.00324	0.00464	0.00470	0.00566	7.93480
	DT	0.01214	0.01148	0.01214	0.01140	0.01311	0.00596	14.18478
FM + llama2	SVM	0.01040	0.00947	0.01040	0.00985	0.01286	0.00545	0.92939
	NB	0.01087	0.02383	0.01087	0.01045	0.01046	0.00561	0.01881
	MLP	0.00981	0.00714	0.00981	0.00839	0.01129	0.00592	3.08061
	KNN	0.00347	0.00317	0.00347	0.00368	0.00584	0.00291	0.02702
	RF	0.00698	0.00756	0.00698	0.00867	0.01043	0.00549	1.78260
	LR	0.00373	0.00360	0.00373	0.00350	0.00496	0.00317	7.17186
	DT	0.00963	0.01328	0.00963	0.01144	0.01266	0.00894	2.83893
FM + llama3	SVM	0.00710	0.00361	0.00710	0.00535	0.01009	0.00308	0.90535
	NB	0.00431	0.00631	0.00431	0.00804	0.00466	0.00174	0.06513
	MLP	0.00525	0.00285	0.00525	0.00376	0.00685	0.00525	1.75191
	KNN	0.00568	0.00720	0.00568	0.00740	0.01364	0.00683	0.02675
	RF	0.00612	0.00756	0.00612	0.00675	0.01533	0.00747	4.20123
	LR	0.00806	0.00635	0.00806	0.00709	0.01328	0.00449	5.71633
	DT	0.01023	0.01087	0.01023	0.01048	0.01679	0.00930	2.80694
FM + mistral	SVM	0.00463	0.00567	0.00463	0.00474	0.00605	0.00612	0.70354
	NB	0.00886	0.01288	0.00886	0.01301	0.00901	0.00511	0.03534
	MLP	0.00752	0.00753	0.00752	0.00707	0.01231	0.00835	3.76857
	KNN	0.00716	0.00856	0.00716	0.00774	0.01159	0.00490	0.03406
	RF	0.00544	0.00637	0.00544	0.00486	0.00866	0.00252	2.53585
	LR	0.00398	0.00582	0.00398	0.00526	0.00846	0.00708	5.67962
	DT	0.00495	0.00253	0.00495	0.00358	0.00798	0.00318	7.91662

Table 5: Standard deviation results for MIMIC-IV dataset without discharge notes.

Embed. Method	ML Algo.	Acc.	Prec.	Recall	F1 weigh.	F1 Macro	ROC-AUC	Train. runtime (sec.)
Classical FM	SVM	0.00862	0.00951	0.00862	0.00897	0.01401	0.00617	0.04517
	NB	0.01033	0.00665	0.01033	0.00865	0.01338	0.00538	0.00121
	MLP	0.01139	0.00950	0.01139	0.01036	0.01623	0.00621	1.27089
	KNN	0.00887	0.01051	0.00887	0.00900	0.01667	0.00708	0.00493
	RF	0.01090	0.01200	0.01090	0.01165	0.02146	0.01174	0.02656
	LR	0.00900	0.01059	0.00900	0.01023	0.01889	0.00954	0.00307
	DT	0.00672	0.00704	0.00672	0.00679	0.01721	0.00924	0.00322
llama2	SVM	0.00391	0.01286	0.00391	0.00822	0.02474	0.01293	1.45777
	NB	0.00728	0.01181	0.00728	0.00773	0.00964	0.00758	0.03113
	MLP	0.00488	0.00898	0.00488	0.00864	0.00735	0.00227	0.42210
	KNN	0.00801	0.01942	0.00801	0.01261	0.00817	0.00170	0.06186
	RF	0.00845	0.01583	0.00845	0.01239	0.00225	0.00000	3.91316
	LR	0.00463	0.01279	0.00463	0.00840	0.01919	0.01093	0.57689
	DT	0.00861	0.01767	0.00861	0.01200	0.01801	0.01098	11.53753
llama3	SVM	0.00468	0.00548	0.00468	0.00524	0.00775	0.00784	0.25689
	NB	0.00296	0.00229	0.00296	0.00276	0.00531	0.00177	0.01825
	MLP	0.00280	0.00275	0.00280	0.00275	0.00613	0.00381	0.39162
	KNN	0.00389	0.00340	0.00389	0.00370	0.00751	0.00348	0.03080
	RF	0.00183	0.00174	0.00183	0.00177	0.00287	0.00156	0.58341
	LR	0.00310	0.00365	0.00310	0.00327	0.00453	0.00407	0.58107
	DT	0.00460	0.00410	0.00460	0.00430	0.01064	0.00406	2.41015
Mistral	SVM	0.00416	0.00437	0.00416	0.00422	0.00875	0.00500	0.29996
	NB	0.00544	0.00346	0.00544	0.00495	0.00884	0.00226	0.01924
	MLP	0.00191	0.00314	0.00191	0.00215	0.00247	0.00481	0.37578
	KNN	0.00278	0.00226	0.00278	0.00260	0.00473	0.00186	0.05112
	RF	0.00372	0.00345	0.00372	0.00357	0.00579	0.00336	1.44371
	LR	0.00466	0.00479	0.00466	0.00463	0.00864	0.00503	1.71386
	DT	0.00614	0.00593	0.00614	0.00602	0.01136	0.00526	3.77916
FM + llama2	SVM	0.00389	0.00428	0.00389	0.00406	0.00538	0.00415	0.66821
	NB	0.00194	0.00185	0.00194	0.00184	0.00434	0.00208	0.02384
	MLP	0.00240	0.00280	0.00240	0.00250	0.00385	0.00393	0.85022
	KNN	0.00462	0.00437	0.00462	0.00451	0.00679	0.00328	0.04094
	RF	0.00252	0.00254	0.00252	0.00245	0.00508	0.00306	1.91991
	LR	0.00301	0.00321	0.00301	0.00302	0.00422	0.00383	1.42755
	DT	0.00635	0.00652	0.00635	0.00650	0.01136	0.00666	4.57725
FM + llama3	SVM	0.00501	0.00514	0.00501	0.00503	0.00796	0.00480	0.33299
	NB	0.00205	0.00141	0.00205	0.00189	0.00220	0.00166	0.00671
	MLP	0.00340	0.00350	0.00340	0.00338	0.00693	0.00508	1.25097
	KNN	0.00194	0.00180	0.00194	0.00187	0.00218	0.00217	0.03127
	RF	0.00354	0.00281	0.00354	0.00334	0.00468	0.00194	0.81135
	LR	0.00466	0.00410	0.00466	0.00446	0.00722	0.00307	0.75116
	DT	0.00578	0.00540	0.00578	0.00560	0.00916	0.00539	3.51961
FM + mistral	SVM	0.00750	0.00836	0.00750	0.00784	0.01013	0.00782	0.35105
	NB	0.00508	0.00318	0.00508	0.00465	0.00730	0.00167	0.01665
	MLP	0.00581	0.00566	0.00581	0.00573	0.00745	0.00387	1.47348
	KNN	0.00504	0.00494	0.00504	0.00493	0.00593	0.00366	0.02254
	RF	0.00312	0.00229	0.00312	0.00292	0.00460	0.00114	2.01679
	LR	0.00622	0.00748	0.00622	0.00650	0.00762	0.00726	0.81341
	DT	0.00673	0.00804	0.00673	0.00739	0.00893	0.00759	1.95065

Table 6: Standard deviation results for MIMIC-IV dataset with discharge notes.